

JONATHAN EARP

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linkedin.com/in/earpjonathan | github.com/earpjonathan | FAA Part 107 Remote Pilot | U.S. Citizen

I've tuned PID loops on eight drones and a VEX robot, using flight logs and test runs to decide each change, and I calibrate the GPS, compass, barometer and IMU on every build. On a 3D reconstruction project I worked out an uncalibrated lens's focal length by regressing optical flow against gyro rotation.

EDUCATION

University of Edinburgh Edinburgh, UK · 2026 – 2031
MEng Electrical and Mechanical Engineering, Year 1

The Head-Royce School Oakland, CA · Jun 2026
GPA 3.80 unweighted, 4.38 weighted. Honors Calculus, Advanced Physics, Advanced Chemistry, Data Science

PROJECTS

FPV and UAS Builds — Personal | *Betaflight, ArduPilot, ExpressLRS* 2023 – Present

- Calibrate GPS, compass, barometer and IMU on every build, and tune PID loops over repeated test flights using flight logs.

3D Reconstruction from FPV Flight Data — S.P.L.O.O.G.E | *Python, COLMAP, Rust, WebGL* Aug 2026

- Built a pipeline from raw DJI O4 Pro video and gyro telemetry to one 3D Gaussian Splatting model of two flights six days apart: 7,470 of 7,471 frames registered at 0.82 px mean reprojection error.
- Estimated the uncalibrated lens focal length by regressing optical flow against gyro rotation ($R^2 = 0.967$), and cut video-to-render misalignment from 72.3 px to 0.89 px by undistorting the footage.

Head-Royce VEX Robotics — Team Captain Oakland, CA · 2022 – 2026

- Designed and built robot mechanisms, then tuned PID control with the programmers over repeated test runs.
- State Champion, Worlds Division Finalist, 2× Tournament Champion, 2× Excellence, 3× Build, 2× Amaze, 1× Design.

Automated Video Rendering Pipeline — P.E.T.E.R | *Python, OpenCV, Playwright, ffmpeg* 2026

EXPERIENCE

Lawrence Berkeley National Laboratory — Student Research Assistant Berkeley, CA · Apr 2024 – Present
StoveCam: low-cost camera sensor for an indoor air quality and childhood asthma study

- Built a Raspberry Pi 5 unit with RGB and thermal cameras that captures the cooktop every 60 s in occupied homes, then reworked the mounts, wiring and capture scripts until it ran reliably across different kitchens.

University of Pennsylvania, Singh Center for Nanotechnology — ESAP Participant Philadelphia, PA · Summer 2025

SKILLS

Controls & sensors: PID tuning, IMU and gyro data, quaternion orientation, sensor calibration, flight-log analysis

Electrical & avionics: Brushless motors, STM32 ESCs, soldering, Betaflight, ArduPilot, ExpressLRS, analog and DJI video

Programming: Python (NumPy, OpenCV, Playwright), Rust, COLMAP, JavaScript, WebGL, Arduino, Raspberry Pi / Linux, Git

CAD & fabrication: Onshape, Fusion 360, 3D printing, CNC, laser cutting, bandsaw, drill press, photolithography

LEADERSHIP

- **Medical Advocacy, President.** Ran a blood drive that brought in 120 donors, and set up a partnership with a local hospital.
- **Eagle Scout, Senior Patrol Leader.** Taught a month-long robotics class at an elementary school for my Eagle project, and left the curriculum behind so someone else could run it again.